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Abstract - Agriculture faces major challenges due to Unpredictable weather conditions, inefficient irrigation, intrusion of animals causing damage to crops, and fire-related accidents are some of the major challenges faced in agriculture. It results in crop damage during the stage of harvesting or drying because of excessive rain. It contributes to poor irrigation practices resulting in water being wasted and reduces production. Animal intrusion and fire hazards pose serious threats related to crop safety.

This project aims to design a smart system for automation of grain drying, rain protection, irrigation, and crop safety using the Internet of Things with Telegram. This system keeps watch of environmental parameters like rain, soil moisture, temperature, humidity, motion, and fire levels with the help of numerous sensors. Under heavy rain, the system maintains crop safety with a protective shield according to soil moisture conditions. When no rain occurs, crop irrigation and grain drying systems get automated as needed. Animal motion sensors sense any animals approaching the crop area, whereas fire sensors sense any fire accidents, sending signals to the buzzer for immediate attention.

All system activities and emergency alerts to farmers are notified in real-time through the Telegram messaging system, which facilitates remote system monitoring.

1. INTRODUCTION

Agricultural production is at the core of many developing countries. For the majority of their population, their main source of income is agricultural production. An individual farmer's profitability depends on his ability to employ efficient agricultural methods and to surmount the obstacles he faces. Some of these obstacles include lack of predictable weather, poor irrigation systems, loss of harvested grains due to poor handling, and crop loss due to wild animals and climatic occurrences such as fire. All these factors contribute to the reduction of the standard of the quality of the crops, which is essential for profitability.

A prime cause for loss of productivity in farmers is when, during the harvesting season, rain falls in large quantity when it is least expected. If moisture is present in the crops or grain, it may encourage the formation of fungus, resulting in a loss of added value for the grain. Previously, farmers resort to covering their crops with plastic sheets. This is a very time-consuming approach, which fails when rain falls suddenly. Therefore, a real-time automated facility to provide rain protection is required for the crops.

2 . LITERATURE SURVEY

Previous research in smart agriculture focused on automated irrigation based on soil moisture sensors for the reduced loss of water. Various rain detection systems were proposed that protect crops but remain independent, not integrated with any irrigation or grain-drying process.

Other systems that were introduced allowed the grain to be dried using temperature and humidity sensors, although these needed manual supervision. Independently, animal detection systems with motion sensors and fire detection systems with flame sensors were developed with no central monitoring or warning mechanism.

The recent IoT- based solutions brought cloud monitoring, but it was quite expensive and cumbersome for the farmers. This calls for the development of an integrated cost-effective, real-time alert-based agricultural system. The proposed system integrates rain protection, irrigation, grain drying, animal intrusion detection, and fire detection with notifications via Telegram in a highly efficient and farmer-friendly manner.

It is just as critical to dry grains after they are harvested as it is to apply the proper method of storing them; grains must be dried down to the optimum moisture level before being stored properly. If grains are improperly dried, they will spoil, can be infested by pests and will be significantly reduced in quality. Manual methods of drying will typically require appropriate weather conditions and will require repetitive supervision by humans, which often is not an option.

1. Automated Grain Drying Systems

Proper drying of grains is a crucial post-harvest management technique. Bhatta charya et al. (2021), in their research, proposed the use of a sensor-based approach for the automatic drying of grains. They used temperature and humidity sensors as inputs for the system. This idea improved grain drying efficiency, but the use of other conditions such as rain and irrigation during the agricultural seasons was not taken into account.

2. Animal Detection and Crop Protection

Animal intrusion is a major concern, which affects crop yield and income for farmers. Some studies used PIR motion sensors to sense the intruders entering the area of the crop farm. For instance, Reedy et al. (2022) proposed the use of a motion detection system for alerting farmers. However, this system lacks integration with other environmental parameters, such as irrigation and rain protection. 3 Fire Detection in Agricultural Systems The fire safety issue in agricultural field areas and storage areas is critical. Chatter e a. (2020) proposed a system using flame sensors and detecting fire with the aim of sending alert messages via SMS. The proposed system improved the security of the smart agricultural system but was isolated from other agricultural services.

3 . PROPOSED SYSTEM

3.1 System Architecture

It consists of an IoT controller (ESP32/Arduino sensors for the environment, actuators, and a notification module for Telegram. Architecture The proposed solution incorporates three major functionalities:

1. Automated grain drying
2. Rain detection and protection
3. Smart irrigation control

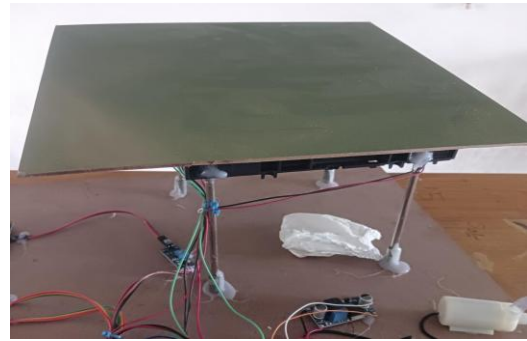
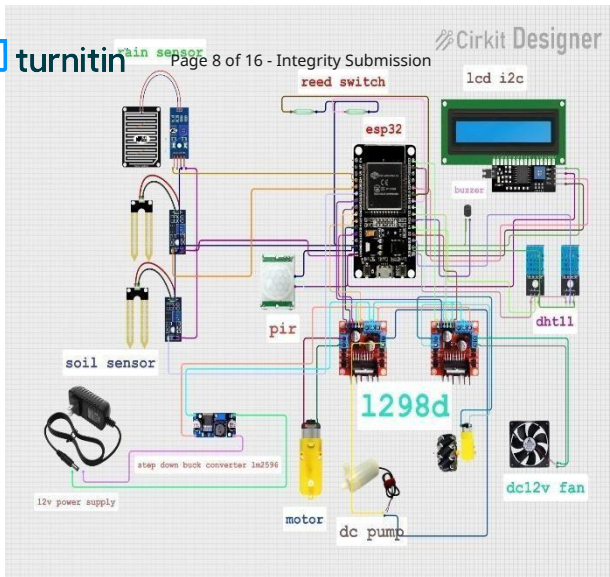


FIG : RAIN SHIELD

The rain shield is a self-operated protection system to save crops and grains from rains accompanied by heavy rainfall.

Main Components:

- ESP32 / Arduino microcontroller
- Temperature and humid sensor
- Rain sensor
- Soil moisture sensor
- Motor/Heater/Fan
- Motorized cover for Rain cover
- Water pump for irrigation
- Relay

Working Principle:

- Telegram Bot for alerts
- Sensors constantly monitor environmental conditions
- ESP32 processes sensor data
- Appropriate actions are taken:
 - Rain → Shield closes
 - No Rain + Low Moisture → Irrigation ON
 - High humidity → Fan ON
 - Animal Movement → Alert
 - Fire → Buzzer + Alert
- System Status displayed on LCD
- Telegram alerts sent to the farmer



FIG : SOIL SENSOR



FIG : DC MOTOR

Used for water pumping whenever it is needed.



FIG : DRAIN DRYER

The grain dryer automatically reduces excess moisture in grains.

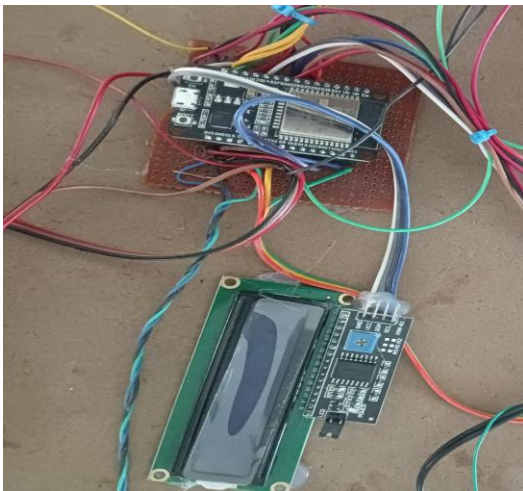


FIG : LCD DISPLAY & ESPE2 MICRO CONTROLLER

LCD: Displays system status and sensor values
 ESP32 Microcontroller: Acts as the main controller and handles sensor data, actuator control, and IoT communication.

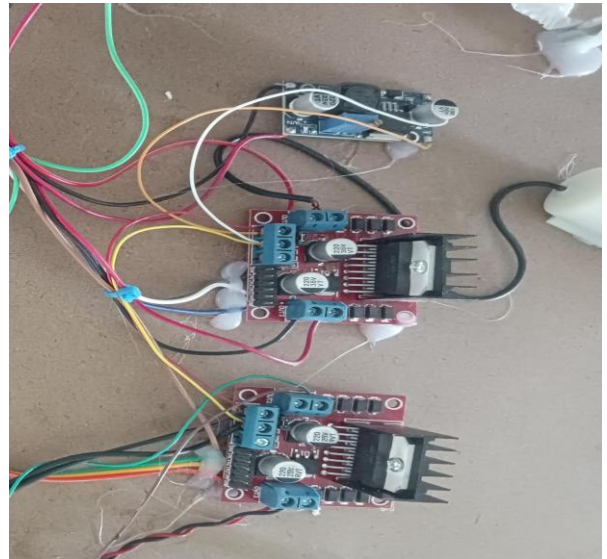


FIG : L298D MOTOR DRIVER MODULE

The L298D motor driver module is used to control the direction and speed of DC motors for operating the rain shield and grain dryer mechanisms by interfacing them with the microcontroller..



FIG : BUZZER

The buzzer provides an audible alert during emergency conditions such as fire detection or animal intrusion, while simultaneously sending notifications to the farmer via Telegram.

4. IMPLEMENTATION DETAILS

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The system continuously reads data from various sensors such as rain, soil moisture, temperature, humidity, motion, and fire. The data is then processed by the microcontroller and compared with certain threshold values.

4

- Shield mechanism is operated by a motor via relay.
- The irrigation pump will be switched ON/OFF based on soil moisture.
- The system used for grain drying normally functions based on humidity.
- Animal intrusion causes motion alerts.
- Fire detection

Telegram Bot API offers "send instant notifications":

- Grain drying system activated
- Animal detected near crops

Fire detected – buzzer ON Hardware Configuration

- ESP32 as central controller
- Sensors linked to digital and analog pins
- Relay Module for Motor, Pump, and Dryer Control

Buzzer for emergency alert Wi-Fi for Telegram communication Software Configuration In

Arduino IDE ☐ Embedded C ☐ Telegram Bot API
Sensor threshold-based logic

3

If any animals enter the area, and fire accidents occur, the system immediately sends alerts to the farmer with the help of PIR sensors and fire sensors, which activate the buzzer. All activities of the system occur with the proper display of the alerts on the LCD by sending immediate Telegram messages to the farmer.

Automating the process of drying grains with monitoring of both temperature and humidity provides consistent and safe drying conditions

The system utilizes ESP32, which acts as a micro controller. When this ESP32 gets various sensor and actuator inputs, it performs different agricultural-related tasks. Rainfall, soil moisture, temperature, humidity, motion, and fire sensors continuously sense various parameters of the crops and grains. Depending upon the inputs from these sensor devices, the ESP32 regulates the protective shield, irrigation pump, grain dryer, and buzzer with the help of various motor drivers. When heavy rainfall occurs, the protective shield covers the crops. Similarly, when no rain is present, irrigation and drying activities occur accordingly. If any animals enter the area, and fire accidents occur, the system immediately sends alerts to the farmer with the help of PIR sensors and fire sensors, which activate the buzzer. All activities of the system occur with the proper display of the alerts on the LCD by sending immediate Telegram messages to the farmer.

The system stresses the need for crop safety and remote monitoring along with environmental automation. The PIR motion sensor detects the motion of animals around the field, while the fire sensor detects fire mishaps in the initial stages. In case of any abnormal or emergency condition, the system will also trigger a buzzer for a local alert with instant notification to the farmer via Telegram using Wi-Fi connectivity of ESP32. In addition, it also displays real-time system status and sensor readings on-site through an LCD display. Such an integrated implementation minimizes human intervention, crop losses, and increases agricultural efficiency based on smart automation.

The system also allows for remote manual control through Telegram commands, which gives farmers flexibility in special situations and can be scaled to multiple units for monitoring large fields through centralized management.

5. RESULTS AND DISCUSSION

The IoT-based automatic grain drying, rain protection, irrigation, and crop safety system developed has been tested under different kinds of environmental conditions. When the rainfall is heavy, it has been found that the rain sensor worked correctly, and the shield has been deployed **based on the moisture level of the soil**, protecting the grains from exposure to rainwater by bringing its deployment process under automatic control. In fact, the Telegram message has been received instantly, signifying that rain has been detected correctly.

The irrigation system worked efficiently because when no rainfall was recognized, the system worked **based on the data** obtained **by the soil moisture sensor**. The water pump worked efficiently by turning ON **when the soil moisture level** went **below the threshold** value **and** turning OFF when the desired moisture level was attained. The grain drying system worked efficiently when there was high humidity; the drying fan worked efficiently to reduce the moisture level surrounding the grains.

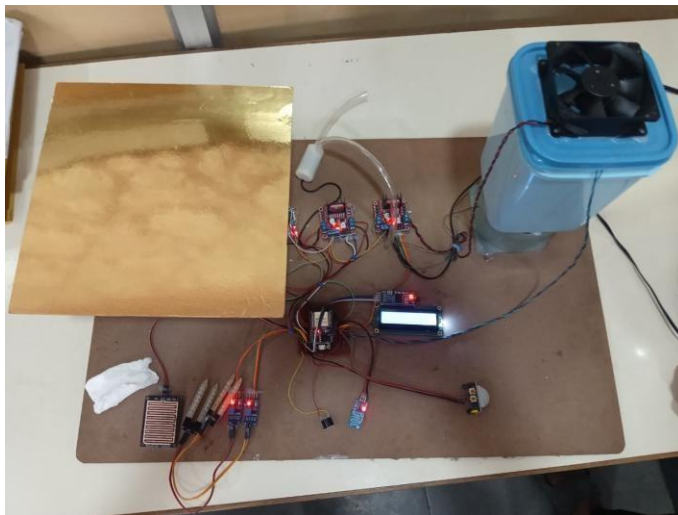


FIG : AGRI SHIELD

From the results obtained, it is clear that the movement of the animal in the vicinity of the targeted crop area has been successfully identified by the PIR sensor and immediate alerts were confronted by the farmer via Telegram. From the results of the fire detection, it is confirmed that the presence of flame has been successfully identified by the sensor and, at the same time, activated the alarm as well as the emergency messages. Hence, the functionality of the proposed system is clear and confirmed to be effective and cost-effective for agriculture.

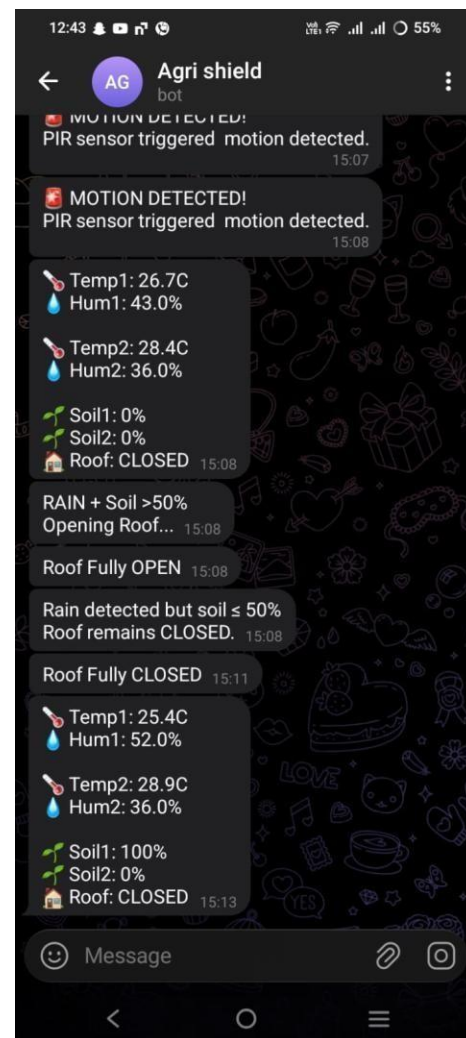


FIG: TELEGRAM NOTIFICATION

6. CONCLUSION

The application of IoT technology for the automated grain drying, rain protection, irrigation, and crop safety system has been a notable success and proves how modern technologies, especially for sensing and communication, can be applied for the betterment of the field of agriculture. It can respond to environmental changes by providing a shield during rain, taking care of irrigation levels according to soil conditions, and drying grain when it is humid. This automation is helpful in curtailing losses. Apart from this, the integration of animal intrusion detection system with fire detection system adds to the safety of the crops. The real-time telegram notification with buzzer signals to the farmer adds to the ease of use for farmers to get timely information about the activities of the system. Overall, the proposed system has promising results for smart farming, which is cost-effective, efficient, and safe.

7. FUTURE WORK

In the future, the system proposed in this thesis may be extended to integrate solar power to enable an energy-efficient system and to accommodate remote agricultural areas where there is limited electricity supply. Weather forecasting API may also be incorporated in the system to be able to control and anticipate rainfall and temperature in advance, controlling the shield, water supply, and grain drying equipment accordingly. A mobile application may be developed to give the farmer graphical user interface and manual control over the sensor readings and equipment operation, including Telegram bot functionality. The rain shield, irrigation, and grain drying systems can be completely automated with AI technology, which will enhance efficiency in their operation and minimize human involvement.

Further improvements can include the use of AI techniques to get the soil, crop, and water requirements for better irrigation scheduling. Instead of motion detection, camera-based animal detection can be used for precise detection. Moreover, the addition of fire suppression systems and large-scale cloud storage for further analysis can make the system more intelligent, reliable, and scalable for smart farming purposes.

Furthermore, voice alert systems and voice-based command systems can also be integrated, specifically in regional languages, such that farmers can also be alerted or can access the system by simply giving voice commands.

Application of GPS and geofencing can also enhance the surveillance of broad agricultural lands and control the extent of animal movement.

Edge computing can be used to process the data from the sensor, ensuring systems are still operational, especially when internet connectivity is poor.

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